Poshan Setu: turning child-growth records into timely, accountable action

The India @71/100 Challenge · Theme 1: Maternal & Early Childhood Nutrition · Round 1 proposal (v0.3, 4 Oct 2026) Team: *[name]* · **Team ID:** *[after registration]* · **Institution:** IIT Delhi *(+ teammates)* · **Contact:** *[email, phone]*

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1. Problem identification: the gap and who falls into it

In May 2026, Poshan Tracker recorded growth monitoring for **over 6.3 crore children aged 0–5** across all States and UTs [3]. Yet the system and the national survey disagree sharply on wasting, the most dangerous form of undernutrition:

Children	NFHS-5 (2019–21, under 5) [1]	Poshan Tracker, Nov 2023 [1]	Poshan Tracker, June 2025 (0–5 y) [2]
Stunted	35.5%	37.51% (under 6)	37.07%
Underweight	32.1%	17.43% (under 6)	15.93%
Wasted	**19.3%**	**6%**	**5.46%**

The divergence is in the weight-based indicators, and it is universal: Poshan Tracker's wasting is below NFHS-5's in **all 36 States/UTs** (e.g. Maharashtra 4.01% vs 25.6%) [2, 14]. The figures aren't directly comparable: the age bands differ, the periods differ, and one source samples households while the other counts children registered and measured. So this is an **unexplained discrepancy**, not proof of missed children. The Ministry itself notes that administrative data "suffer from data gaps due to coverage, measurement, training etc." (Rajya Sabha reply, as reported in [6]). Today no one can say, for any district or Anganwadi, how much of the gap is real change, coverage, measurement error or misclassification.

Who falls into the gap. Infants and young children (0–23 months) in high-burden, tribal-majority blocks. - Wasting starts early: in South Asian cohorts, 18.9% of babies were wasted at birth and 52.2% had at least one episode by age two [7]. - Reporting from Melghat (Maharashtra) describes infant deaths and coordination failures alongside substantial recorded malnutrition (1,82,443 malnourished children statewide in Feb 2025) [5]. It does not establish that the children who died had been identified and followed up. - Seasonal migrants and unregistered children may never be measured at all.

The measurable gap we target, locally and repeatably: 1. The share of new wasting episodes preceded by an actionable flag at least 30 days before onset. 2. The days from first actionable flag to action: a home visit, or medical appraisal for SAM. 3. Agreement between records and blinded re-measurement.

The pilot question: *do existing measurements lead to a named owner, timely assessment and completed care, and which children never enter the records at all?*

2. Evidence & context

- **Scale and tools exist:** >8.93 crore registered beneficiaries; 13.97 lakh Anganwadi centres (AWCs). Every Anganwadi worker (AWW) has a smartphone and every AWC has growth-monitoring devices. Poshan Tracker classifies each measurement against WHO standards and schedules 23 home visits from pregnancy to age 3 [3, 4].
- **Speed matters:** in the pooled ki cohorts, 71.1% of wasting episodes ended within 90 days, fewer in South Asia [7]. Monthly data only help if acted on within weeks.
- **Another app isn't the answer:** the CommCare-based ICDS-CAS raised adequate home visits by 7.9–8.3 percentage points and counselling recall by 8–11.5 pp (Bihar, MP), but had "no effects … on infant and young child feeding practices" [8]. Workers

already report duplicate paper and app registers, device and network failures, and exclusion at the face-authentication step for Take-Home Ration [6].

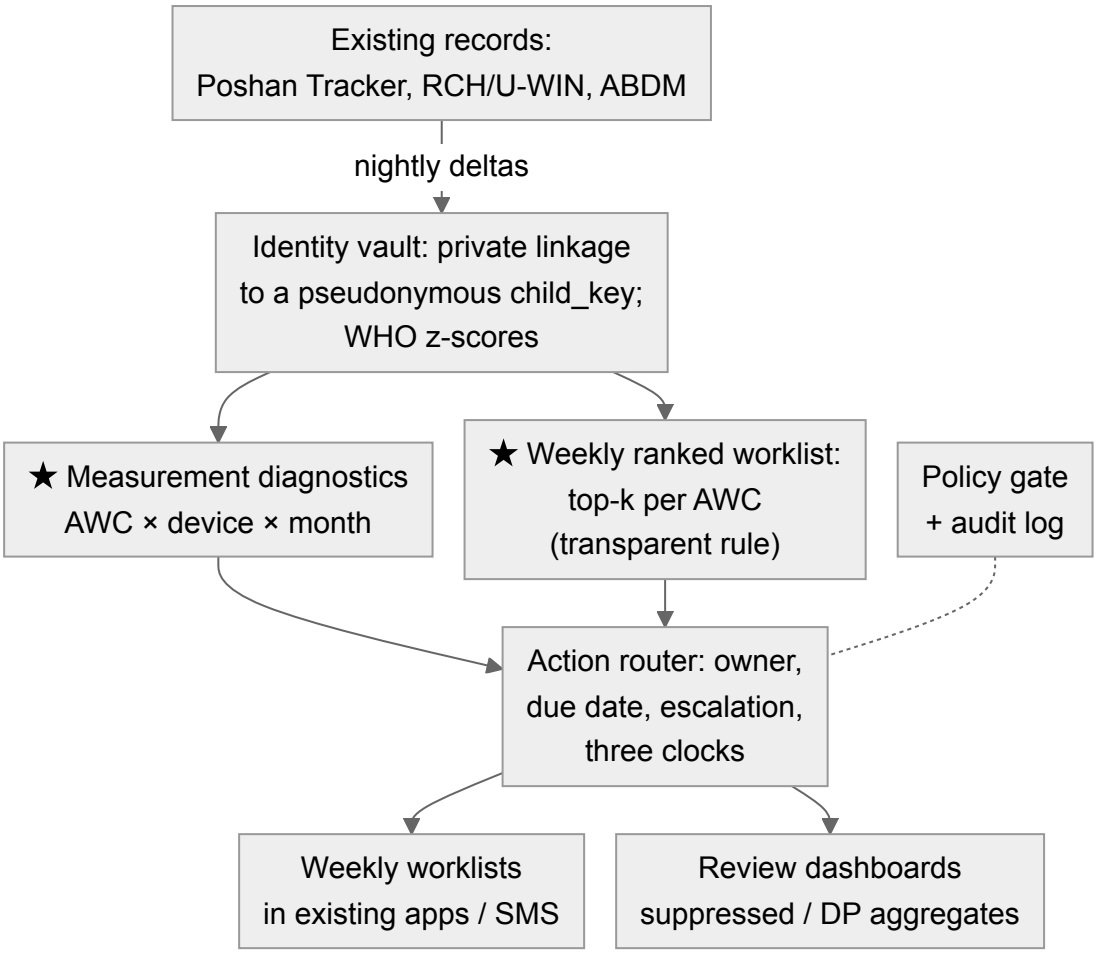
- **States run campaigns, not continuous governance:** UP's Sambhav Abhiyan targets the 100 highest-stunting AWCs per district for monthly committee review [9].

3. Existing ecosystem: why the gap persists

Layer	What exists	Not in Poshan Tracker's published feature set [3], so our target
Measurement	App, devices, WHO classification	A check on the measurements themselves: heaping, copy-forward, implausible trajectories
Interoperability	RCH/ANMOL and U-WIN hold birth weight, illness, immunisation; ABDM consent-based exchange [10]	A routine link to Poshan Tracker records; names spelled differently; IDs often absent
Decision support	Point-in-time SAM/MAM status; review dashboards	Weekly lists sized to each worker's capacity
Accountability	Sector, project and district reviews; national CMAM protocol timelines [11]	An owner, due date and escalation clock per alert; a record of who saw what
Governance	DPDP Act 2023 and Rules 2025; ABDM consent manager	Purpose limits enforced in code; disclosure control on published aggregates

4. Proposed solution: Poshan Setu, a governance and decision layer behind existing apps

Poshan Setu **reuses the growth records the system already collects**. Each week it decides what is credible and what is urgent, and returns a short worklist into the apps workers already use. Worklist delivery and acknowledgements depend on agreed departmental interfaces. Every data access passes a purpose-bound policy check and is written to a tamper-evident log.



4.1 Interoperability and data flow (sub-area 1). - Nightly deltas from Poshan Tracker and RCH/U-WIN, plus ABDM health records under consent artefacts that fix purpose and duration [10], land in a state **identity vault**. - Records are linked on shared IDs where they exist. Otherwise we use probabilistic matching tuned for Indian names ("Lakshmi/Laxmi", "Baby of Sunita", DOB recorded as the 1st), or keyed Bloom-filter encodings so departments can link without exchanging names. - Only a pseudonymous child_key leaves the vault. The SAM list reaches the ASHA/ANM within the protocol's 2 days [11]. - **Minimum deployment contract:** source record and version, child reference, action, owner, due date, blocked-input status, acknowledgement. These fields come from existing exports and the worker's own app.

4.2 Capacity-capped weekly worklists (spike B). - Children observed as SAM/MAM (WHZ, MUAC at 6–59 months, oedema) go straight to the national protocol pathway [11]. - Every AWC also gets *K* slots a week, set to its worker's visit capacity, filled with the children closest to the MAM line. An AWW can check that rule herself. - In our leakage-free simulation, this rule flagged 43% of new wasting episodes a month or more before onset, against 37% for the programme's "no weight gain in 2 months" rule and 27% for random. Our ML model (38%) did not beat it, and linked health records added no predictive gain. So the rule ships first; a model replaces it only if it does better on pilot data. Linkage is justified by care coordination, not prediction. - WHO growth-velocity scores are used only within their published range (to 24 months). - **Legal go/no-go:** each identifiable data flow needs a written legal basis that covers consent *and* any restriction or exemption. Without it, deployment is limited to lawfully supplied, non-identifying aggregates.

4.3 Data quality and governance safeguards (spike A, sub-area 3). - *Record checks:* WHO plausibility flags and impossible trajectories, such as height falling more than 1.5 cm or three identical weights in a row. - *Measurement diagnostics* per **AWC × device × month, never per worker**: digit preference, date-of-birth heaping, the spread of WHZ (expected near 1 SD), missingness, and trajectory realism (copied-forward data rarely dips the way real measurements do). - *Support before scrutiny:* a flag first triggers an equipment calibration and a stock check. The Lady Supervisor re-measures a sample of children in flagged AWCs *and* in randomly chosen unflagged AWCs, with known selection probabilities. - Workers see the same diagnostics, can annotate them ("scale broken since the 12th") and can appeal. Remote and migrant-serving AWCs are compared only with similar AWCs. - HR, pay, transfer or disciplinary use is denied in the policy code and the data-sharing agreement. Residual misuse risk is covered by oversight and grievance. - **The law:** - Every beneficiary is a "child" under the DPDP Act 2023 (s.2(f)); ss.7–10 commence on **13 May 2027** (G.S.R. 843(E)) [12]. - The State must, "through the local anganwadi, identify" children who suffer from malnutrition (**NFSA 2013 s.6**) [13]. Processing for that function (DPDP s.7(c)) "in the interests of a child, under any law" is exempt from the parental-consent and no-tracking rules (DPDP Rules 2025, Rule 12, Fourth Schedule B(1)) [12]. - s.8(3) requires "completeness, accuracy and consistency" when data drive decisions. *The diagnostics are how the State meets that duty.* - Research is allowed only without individual decisions (s.17(2)(b)). Public figures use small-cell suppression or differential privacy.

4.4 Decision rights and accountability (sub-area 2). | Role | Sees | Must act (source) | Escalates to | |---|---|---|---| | AWW | Named children of her AWC, with reasons | Share the SAM list with ASHA/ANM within 2 days; MAM fortnightly visits [11] | Lady Supervisor (LS) at the weekly review | | ASHA / ANM / MO-PHC | A named child only with an open joint task | Medical appraisal within 3–5 days of the appetite test [11] | CDPO + MO-PHC (our day-14 trigger) | | LS | Pseudonymous sector view; names only for escalated tasks | Weekly village review [11]; re-measurement sample | CDPO | | DPO / District Magistrate | Suppressed aggregates, delays, input gaps | Monthly district review, especially SAM [11] | State quarterly review [11] |

Three clocks, so no child disappears: - The *child's clock* runs from the first actionable flag and is always visible. - Missing inputs (scale, rations, NRC bed, transport) start a separate *resource clock*, escalated to CDPO/DPO. They never pause the child's clock or an urgent assessment. - SAM, infants under 6 months and danger signs bypass ranking. - A *worker clock* starts only once the inputs exist. Every escalation and data access is logged.

5. Working prototype (what already runs)

Every number below comes from a committed run in our repository. **SYNTHETIC rows are engineering checks on a simulated district, not validation.** The simulator is parameterised from published cohort estimates but does not fully reproduce them: at-birth wasting is 22.8% against 18.9% [7].

Component	Check	Result
WHO z-scores (real WHO tables)	Max error vs WHO's published tables, 87,408 points at ±4 SD	**0.003 z**
Record linkage (SYNTHETIC, 5,000 children)	F1: shared IDs only / probabilistic / privacy-preserving Bloom filter	**0.51 / 0.94 / 0.97**
Measurement diagnostics (SYNTHETIC, 120 AWCs, 3 seeds)	AUROC: any unreliable AWC / copy-forward AWCs. A tuning change that hit	**1.00 / 0.69**

Component	Check	Result
	0.95 failed a different fabrication pattern and was rejected	
Weekly worklists (SYNTHETIC, 10,690 new episodes, 5 slots per AWC per month)	Episodes flagged ≥1 month ahead: WHZ rule (95% CI) / ML / programme rule / random	**0.43 (0.42–0.44) / 0.38 / 0.37 / 0.27**
End-to-end weekly run (SYNTHETIC, 120 AWCs, 5,307 children)	Tasks → worklists; SMS length; escalated after 8 days (70% on time assumed); audit chain	**1,685 → 140; ≤80 chars; 482; verified**
Policy gate / audit log	Red-team cases (spoofed scope, fake consent, employment use) / tampering incl. truncation	**39/39 / 100/100**

6. Feasibility and implementation roadmap

- **Nov–Dec 2026:** prototype hardening; finals demo.
- **Before any pilot:**
- Paid co-design with Anganwadi worker unions and AWWs from tribal, migrant and low-connectivity areas.
- A written protocol on prohibited uses, workload, correction and grievance, with worker seats in pilot governance.
- A legal opinion for each data flow; a pre-registered evaluation.
- **Pilot, months 1–6** (one high-burden district; children 0–23 months; read-only extracts under a data-sharing agreement):
- Household enumeration plus concurrent blinded re-measurement, to estimate *current local* coverage and measurement/classification error. This cannot explain past national change.
- Prospective validation of the ranking under standard care.
- Stepped-wedge rollout by sector, with independently measured outcomes. Sample size to be set with the partner.
- **Team (our plan):** a data engineer, an analyst, a part-time legal/governance lead, a district coordinator, plus state WCD and health nodal officers. Costing to be done with the partner.
- **Compute:** monthly batch jobs. Our synthetic district runs end to end in seconds on a laptop; state scale is to be benchmarked.
- **Scale, months 6–24:** the components are reusable nationally. Expansion depends on state approvals, interfaces, languages, service capacity and pilot evidence.

Risks: - No data access or no legal basis → aggregates-only mode. - Punitive use → no worker-level outputs; prohibited by policy and contract. - Missed migrant and unregistered children → household enumeration, missed-measurement alerts, subgroup reporting.

7. Expected impact and scalability

- **Beneficiaries:** children 0–23 months in high-burden, tribal-majority blocks first, then all Anganwadi children under 6. Workers get fewer, better-targeted tasks.
- **Pilot outcomes:**
- share of new wasting episodes flagged ≥30 days before onset;
- median days from flag to home visit;
- share of SAM appraised within 3–5 days;

- record vs blinded re-measurement agreement, including false-flag rates in remote AWCs;
- worker time and reported burden, which must not rise;
- completed treatment, not just closed tasks.
- **Scale path:** the same standards, hierarchy and app in every state. Each state supplies its policy file, timelines and interfaces.

References (accessed 4 Oct 2026): 1. PIB (MWCD), Rajya Sabha reply, 20 Dec 2023, PRID 1988614. 2. Rajya Sabha Unstarred Q. 477, 23 Jul 2025, Annexure I. 3. PIB Research, "Poshan Tracker: India's Real-Time Nutrition Monitoring Platform", 8 Jul 2026. 4. Digital India (MeitY), "Poshan Tracker". 5. *The Hindu*, "Why do infant deaths persist in … Melghat", 2025. 6. *News18*, 5 Dec 2025 (Rajya Sabha reply on Poshan Tracker vs NFHS-5); *Hindustan Times*, 18 Jun 2025 (face-recognition rule for Take-Home Ration); BehanBox, 14 May 2025. 7. Mertens et al., *Nature* 2023, "Child wasting and concurrent stunting in LMICs" (PMC10511327). 8. Patil et al., *BMJ Global Health* 2022;6(Suppl 5):e007298. 9. *Hindustan Times* (Lucknow), "Sambhav Abhiyan 5.0", 2025. 10. ABDM, Guidelines for HIPs, HRPs, HIUs and PHR apps, 24 Oct 2022, pp. 5, 19. 11. MWCD & MoHFW, Protocol for Management of Malnutrition in Children, 10 Oct 2023, pp. 8, 11, 20–22. 12. DPDP Act 2023; G.S.R. 843(E) and DPDP Rules 2025 (G.S.R. 846(E)), 13 Nov 2025. 13. National Food Security Act 2013 (No. 20 of 2013), s.6. 14. Lok Sabha Unstarred Q. 2668, 4 Aug 2023, Annexure (NFHS-5 State/UT-wise).